Engineering Evidence Record

Vulcan

2026-10-15T09:20:43+00:00

Status: independently accepted at report generation.

This model-free evidence record is not an independently reviewed technical product narrative. It was generated after engineering acceptance and retains original administrative records. It describes the admitted scope, not manufacturing or physical signoff. Source line counts measure size, not quality or correctness.

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1 Product objective and admitted specification

Design the first chip in our Anvil AI silicon family (Anvil-E embedding series): Anvil-E1.

1. Product & Authoritative Model Specification (Pure-Text Scope Only): - Target Model: Official HuggingFace google/embeddinggemma-2 —Generation 1 (Anvil-E1) implements the **Pure-Text Edition (embedding_gemma2_text + Pooling + Normalize)** only. Vision and audio modalities are excluded from Gen-1 and reserved for future Anvil-E iterations. - Authoritative HuggingFace pure-text specification files are already provided in ./hf_embeddinggemma2_spec/ (do NOT pause or ask for operator clarification) : * ./hf_embeddinggemma2_spec/config.json: Pure-text text_config (24 layers, vocab_size = 262144, hidden_size = 512, hidden_size_per_layer_input = 512, intermediate_size = 2048, embedding_dim = 768, hidden_activation = "gelu_pytorch_tanh", rms_norm_eps = 1e-6, 5:1 hybrid attention schedule across 20 sliding_attention layers [sliding_window = 512, 4Q:2KV, head_dim = 256, rope_theta = 10000.0] and 4 full_attention layers at indices 5, 11, 17, 23 [4Q:1KV, head_dim = 512, rope_theta = 1000000.0]). * ./hf_embeddinggemma2_spec/language_model_tensors.json: Exact pure-text language_ model.* tensor manifest and shapes extracted from HuggingFace google/embeddinggemma-2/resolve/ main/model.safetensors (embed_tokens [262144, 512], ple.per_layer_model_projection [12288, 512] , ple.per_layer_projection_norm [512], per-layer input_layernorm, self_attn. {q,k,v,o}_proj, self_attn. {q,k}_norm, post_attention_layernorm, pre_feedforward_layernorm, mlp. {gate,up,down}_proj, post_ feedforward_layernorm, ple_block. {per_layer_input_gate,per_layer_projection,post_per_layer_input_ norm}, layer_scalar, final norm [512], and embedding_projection [768, 512]). * ./hf_embeddinggemma2_ spec/modules.json, ./hf_embeddinggemma2_spec/1_Pooling/config.json, and ./hf_embeddinggemma2_ spec/sentence_bert_config.json: Sentence-Transformers pipeline (0_Transformer -> 1_Pooling masked mean pooling -> 2_Normalize L2 normalization with Matryoshka truncation support for 768 / 512 / 256 / 128).

2. Autonomous Architecture, Modular Multi-File RTL & Independent Verification Requirements: - You (Vulcan) have full autonomy to analyze the google/embeddinggemma-2 specification and decide the chip microarchitecture, module decomposition, numerical precision, memory/bus interfaces, design documentation (design/*.md), and verification/synthesis strategy (tests/*). - Multi-file modular RTL discipline: Decompose the hardware architecture into clean, standalone synthesizable SystemVerilog modules under rtl/*.sv with top-level integration module rtl/anvil_e1_top.sv. Put each hardware submodule in its own dedicated rtl/<module>.sv file (one module per .sv file, all listed in design_inputs) rather than packing multiple module definitions into rtl/anvil_e1_top.sv. - Parameterize anvil_e1_top with the exact HuggingFace google/embeddinggemma-2 architectural defaults while supporting reduced parameter overrides so iverilog/vvp simulation and yosys synthesis checks complete rapidly and deterministically within container runtime budgets. - Strict separation of duties: Designer implements the modular RTL (rtl/ *.sv) and design spec (design/*.md), runs at most a fast static syntax/elaboration check (iverilog -g2012 -t null), and immediately returns ready to hand off to Verifier; Verifier independently builds and runs the golden reference model, self-checking testbench, functional simulation, and Yosys synthesis checks.

1.1 Architecture and interface contract

CURRENT DELIVERABLE AND OWNERSHIP Design Anvil-E1 as a single-clock, serialized, external-memory inference accelerator implementing the entire pure-text transformer, not merely a MAC engine, attention scheduler or pooling demonstration. Designer owns rtl/*.sv, design/*.md and design/pack_ model.py. Verifier independently owns tests/*.py and tests/*.sv, including the executable references, stimulus, scoreboard, numerical tests, negative control and synthesis assertions. The supplied hf_ embeddinggemma2_spec files are immutable authoritative inputs. Do not use .vulcan/, .git/, .tooling/ or VULCAN/ as product inputs or outputs. No publication, deployment, tool installation or physical-signoff claim is authorized. Designer may run only a bounded iverilog -g2012 -t null syntax/elaboration check, then immediately hand off to Verifier; Designer must not create or execute functional or synthesis preparation harnesses. Reviewer independently evaluates the admitted plan and actual results. If work cannot finish within a role budget, retain owned sources and return the precise unfinished work rather than claiming readiness.

AUTHORITATIVE ARCHITECTURE Top module is anvil_e1_top in rtl/anvil_e1_top.sv. Expose parameters VOCAB_SIZE=262144, HIDDEN_SIZE=512, PER_LAYER_INPUT_SIZE=512, INTERMEDIATE_SIZE=2048, EMBEDDING_DIM=768, NUM_LAYERS=24, NUM_Q_HEADS=4, LOCAL_KV_HEADS=2, FULL_KV_HEADS=1, LOCAL_HEAD_DIM=256, FULL_HEAD_DIM=512, SLIDING_WINDOW=512 and MAX_SEQ_LEN=262144. Every sixth layer is full attention, beginning at zero-based index 5; the default full indices are exactly 5,11,17,23. Local and full RoPE bases are respectively 10000 and 1000000. Preserve attention_bias=false, inference dropout=0, gelu_pytorch_tanh and RMS epsilon=1e-6. BOS=2, EOS=1 and PAD=0 are tokenizer conventions, not automatic insertion or implicit masking by RTL. The host supplies already-tokenized IDs, absolute positions and explicit token-valid masks; include_prompt=true means no automatic exclusion of prompt tokens. No tokenizer, vision or audio hardware is required.

Expand the manifest's five global tensors and two layer templates into all 413 default pure-text tensors. Every tensor must participate in the actual inference datapath: embed_tokens; per-layer model projection and its projection norm; each layer's four residual-path norms, q/k/v/o projections, q/k head norms, three MLP projections, three PLE-block tensors and layer_scalar; final norm and embedding projection. Local projection widths are Q=4*256 and K/V=2*256; full widths are Q=4*512 and K/V=1*512. Do not incorrectly force either attention output width to HIDDEN_SIZE before o_proj. For reduced configurations, derive PLE projection shapes as [NUM_LAYERS*PER_LAYER_INPUT_SIZE,HIDDEN_SIZE], gate as [PER_LAYER_INPUT_SIZE,HIDDEN_SIZE], and PLE output projection as [HIDDEN_SIZE,PER_LAYER_INPUT_SIZE]. Architectural defaults must be genuine top-level defaults, not documentation-only constants.

MICROARCHITECTURE Use one reusable scalar multiply/accumulate datapath, reusable nonlinear and reduction units, a hardwired tensor-operation sequencer, and external storage for model weights and sequence-sized intermediates. This deliberately prioritizes complete, reproducible functionality over throughput. Use explicit RTL loops/FSM counters, not simulation-only real arithmetic, DPI, a software inference callback, a host-supplied result, or an opaque externally executed operation. The host may pack weights and initialize memory but must not compute hidden states, attention, nonlinearities or final embeddings for the DUT.

Each hardware module has exactly one dedicated rtl/<module>.sv file. Required functional decomposition is anvil_e1_top, anvil_e1_controller, anvil_e1_memory, anvil_e1_mac, anvil_e1_nonlinear, anvil_e1_norm, anvil_e1_attention and anvil_e1_pool. Add further dedicated modules only if implementation clarity requires them, and include every RTL file in design inputs. Integrate each required unit into the live datapath; unused decorative instances do not satisfy modularity. Internal memories must be bounded scalar/vector working storage, not default-size arrays containing the complete checkpoint or MAX_SEQ_LEN squared scores. Attention may repeatedly access external K/V data and compute one query/head at a time. Recompute scores for normalization if necessary to avoid an unbounded on-chip score buffer. Document external-memory traffic and scratch-space sizing without inventing measured performance.

TOP-LEVEL INTERFACE Use clk and active-low asynchronous rst_n. Job inputs are job_valid, job_ready, job_tokens[31:0], job_output_dim[15:0], weights_base[63:0], tokens_base[63:0], positions_base[63:0], mask_base[63:0] and scratch_base[63:0]. Addresses are byte addresses. Accept a job only on job_valid && job_ready; latch all descriptors at that edge. Token IDs, positions and masks each occupy one little-endian 32-bit memory word; masks must be 0 or 1. Positions must be less than MAX_SEQ_LEN. The packer documents token order and scratch layout, and hardware must reject invalid IDs, positions, masks, descriptor dimensions or address alignment before consuming them as valid data. All descriptor regions must be nonoverlapping by the documented host contract; reject address arithmetic overflow in hardware.

External memory uses one outstanding transaction: mem_req_valid, mem_req_ready, mem_req_write, mem_req_addr[63:0], mem_req_wdata[31:0], mem_rsp_valid, mem_rsp_ready, mem_rsp_rdata[31:0] and mem_rsp_error. Every accepted request, including a write, receives one response; responses are ordered and may stall arbitrarily. Requests and their payloads remain stable until accepted. No response is consumed without an outstanding request. Memory errors abort the job and produce an explicit error completion. Successful embedding output uses emb_valid, emb_ready, signed emb_data[31:0], emb_index[15:0] and emb_last. Completion uses done_valid, done_ready and done_error[3:0]; busy is also an output. Define errors as 0=success, 1=invalid descriptor, 2=invalid token/position/mask, 3=memory error, 4=all tokens masked and

5=address overflow. Invalid supported-parameter combinations must be rejected explicitly at elaboration or by a documented parameter diagnostic; never silently clamp them.

During asserted reset, job_ready, mem_req_valid, mem_rsp_ready, emb_valid, done_valid and busy are zero; all payload/status outputs are zero. Reset discards active work and pending output. The memory fabric must share reset or flush old responses before reset release; document that integration requirement. After reset release the idle controller asserts job_ready. A job remains busy through output retirement and completion retirement, and no new job is accepted before done_valid && done_ready. An error produces no success embedding. All-masked input produces error 4 and no embedding. Output data, index, last and completion status remain stable while stalled. Memory requests, embedding words and completion each have an explicit registered holding stage; no combinational path may depend on downstream ready to change a presented payload. Document the actual per-operation FSM latency and register boundaries; overall inference latency is variable with dimensions and memory stalls, not a fixed-cycle pipeline claim.

NUMERICAL CONTRACT Use signed 32-bit Q8.24 values for weights, activations and embedding output. Quantize finite BF16 weights using round-to-nearest, ties-to-even, with saturation to the representable Q8.24 interval; reject NaN and infinity. Use signed 64-bit products and at least signed 96-bit accumulators. Accumulate complete dot products/reductions at their documented precision before rounding and saturating at a defined operation boundary; do not truncate each MAC independently. Preserve signed arithmetic and explicit bit growth. Epsilon is the nearest representable positive Q8.24 value to 1e-6. Zero norms and empty reductions must have defined handling, not division by zero or unknown outputs.

Implement synthesizable GELU(tanh approximation), exp, sin/cos, reciprocal and inverse-square-root using bounded iterative or table/polynomial arithmetic. Designer documents the exact range reduction, iteration/table constants, rounding and saturation behavior. Minimum primitive accuracy against an independent high-precision reference is: GELU absolute error <=2e-4 for |x|<=8; exp absolute error <= 2e-6 on [-16,0]; sin/cos absolute error <=2e-5 after range reduction; reciprocal and inverse-square-root relative error <=2e-4 where their mathematical result is representable and nonzero. Softmax subtracts the maximum before exponentiation; values below -16 may be clamped to zero with the approximation and its error documented. Test endpoints, intermediate points and saturation separately. RMS denominators and normalization divisors must remain positive through adequate intermediate precision. These accuracy limits are acceptance gates, not presumed achievements.

FORWARD CONTRACT AND MODEL TRACEABILITY Document explicit forward equations and tensor-to-operation bindings before implementing the sequencer. The supplied JSON files establish architecture, shapes and the sentence-transformer pipeline, but do not contain executable forward code; distinguish adopted Gemma-family forward conventions from facts directly evidenced by those files. Do not describe synthetic, quantized-reference testing as official BF16-checkpoint parity.

The implemented reference contract is: scale token embeddings by sqrt(HIDDEN_SIZE); project the token embeddings through the global PLE model projection with its stated dimensional scaling, reshape into a distinct PER_LAYER_INPUT_SIZE vector for each layer, and apply the projection norm. Each layer executes input RMSNorm, q/k/v projection, per-head q/k RMSNorm, RoPE, grouped-query attention, o projection, post-attention RMSNorm and residual addition; then pre-feedforward RMSNorm, GELU(gate projection) multiplied elementwise by up projection, down projection, post-feedforward RMSNorm and residual addition; then the PLE branch using GELU(per_layer_input_gate applied to the current hidden state), elementwise multiplication by that layer's projected PLE input, per_layer_projection, post_per_layer_input_norm and residual addition; finally apply that layer's scalar. Use Gemma-style unit-offset RMSNorm weights consistently and state the global PLE dimensional scaling explicitly as division by sqrt(HIDDEN_SIZE) before projection normalization. Final hidden states pass through final RMSNorm and the [EMBEDDING_DIM,HIDDEN_SIZE] embedding projection.

Attention is bidirectional embedding attention, not autoregressive generation. Valid keys are selected by the explicit token mask. Full layers see every valid key. Local layers see valid keys whose absolute position difference from the query is strictly less than SLIDING_WINDOW. Map each contiguous group of NUM_Q_HEADS/KV_HEADS query heads to its KV head, scale scores by 1/sqrt(head_dim), and use the distinct RoPE bases and dimensions for local versus full layers. Document these mask and forward conventions explicitly, because shape manifests alone cannot prove them. The hardware must support arbitrary valid absolute positions, including nonzero positions; do not substitute table indices that only work at position

zero. Masked queries need not contribute to pooling; prevent an empty attention reduction from producing invalid values.

Apply masked mean pooling over projected token embeddings, including valid prompt tokens. For Matryoshka output, select the leading requested channels and L2-normalize that selected vector, rather than truncating an already-normalized 768-vector without renormalization. Default legal output dimensions are exactly 768,512,256,128. For reduced EMBEDDING_DIM divisible by six, the corresponding legal dimensions are E,2E/3,E/3,E/6. A nonempty pooled zero vector yields a successful all-zero vector, with no reciprocal-of-zero operation. Nonzero output should have norm within 0.002 of one.

PACKING AND INTEGRATION DOCUMENTATION Provide design/pack_model.py using Python standard-library facilities, with a documented CLI accepting an externally supplied safetensors file and producing the Q8.24 memory image and tensor address inventory. It must validate names, BF16 dtype and shapes against the expanded supplied manifest, reject missing/unexpected pure-text tensors or malformed files, and never fabricate pretrained weights. Unrelated non-language-model tensors in a multimodal checkpoint may be excluded explicitly. Use deterministic packing: the five global tensor names in lexical order, followed by ascending numeric layer index and lexical tensor suffix; matrices are row-major, each element occupies four bytes, and offsets are relative to weights_base. The RTL address generator and the independently written verification address model must agree with this documented format. Document scratch offsets and sufficient sizes, job submission, memory/reset obligations, errors, precision, parameter restrictions and reproduction commands in design/*.md. Required checkpoint data are not present in this repository; synthetic fixtures are sufficient to test the implementation and packer, but do not establish pretrained embedding quality.

INDEPENDENT VERIFICATION AND ACCEPTANCE Verifier authors the reference directly from documented equations and supplied manifests, without importing Designer arithmetic helpers or reading DUT internal results as expected values. Maintain two independent comparisons: a high-precision mathematical reference on quantized input weights for numerical accuracy, and a separately implemented fixed-point operation-boundary reference for exact transport, indexing and documented rounding. The reference may use Python standard-library decimal/math and integer arithmetic; no additional package installation is required. Drive and observe only public interfaces in the end-to-end scoreboard. Treat any active X/Z, mismatch, missing output, duplicate output, stale response, unexpected error, child timeout or nonzero tool exit as failure. A simulator exit of zero is not sufficient without all assertions and expected transactions completing.

Run seeds 17,89,203 for each of three reduced configurations: A=(V=16,H=4,P=4,I=8,E=6,L=6,Q=2, localKV=1,fullKV=1,localD=2,fullD=4,window=2,maxSeq=8); B=(V=20,H=6,P=4,I=10,E=12,L=12,Q= 4,localKV=2,fullKV=1,localD=2,fullD=4,window=3,maxSeq=8); C=(V=16,H=4,P=4,I=8,E=6,L=24,Q= 4,localKV=2,fullKV=1,localD=2,fullD=4,window=2,maxSeq=4). This is a mandatory nine-run matrix, with multiple jobs per run, and includes independent PLE width, non-power-of-two dimensions, both head geometries and all four default full-layer positions. Exercise all supported output dimensions; one-token and maximum configured sequence lengths; mixed, trailing and all-zero masks; zero pooled vectors; nonuniform weights/inputs and head-distinguishing values; local-window boundaries; nonzero positions; back-to-back jobs; input, memory, output and completion stalls; reset during requests/computation/output; memory failures; invalid descriptors, token IDs, positions and masks; and address overflow. Use enough nonzero, channel-distinct fixture data that attention is not uniformly weighted and every tensor family affects computation. Assert coverage of every operation family and local/full layer execution, not merely final outputs.

For finite nonsaturating fixtures, require exact agreement with the independent fixed-point contract and maximum absolute normalized-output error <=0.02 against the high-precision quantized-input reference. For nonzero reference outputs with pre-normalization norm >=0.01, additionally require cosine similarity >= 0.999. Assert output norm separately. Exercise saturation fixtures under their exact documented fixed-point semantics, not by excluding them from testing. Verify primitive accuracy bounds independently of end-to-end normalization.

VERIFIER-OWNED SCOREBOARD NEGATIVE CONTROL The behavior acceptance runner must execute a disposable negative control through the same public-interface transaction parser and checking path used for DUT results, with the same independent expected results and exact-comparison assertions. First require that an unmodified, complete successful DUT trace passes that path. Make a separate private copy

of this trace and change exactly one accepted emb_data word by one Q8.24 least-significant bit, preserving a legal signed 32-bit value, all handshakes, indices, last markers, completion and remaining transactions. Require the actual scoreboard to reject this copy at the altered embedding transaction with the specific exact-data mismatch and recorded expected/observed values. A fabricated rejection message, alternate mock checker, parser failure, missing prerequisite, unrelated exception, X/Z or timeout does not satisfy this control. The behavior runner fails if the corrupted trace is accepted, if the unmodified trace fails, or if the intended mismatch rejection is not established. An expected rejection is handled only by this narrowly scoped negative-control driver; ordinary DUT mismatches and nonzero exits remain failures. Retain the original trace identity, disposable altered trace, exact mutation location, checking command or entrypoint, raw diagnostics and expected-rejection summary separately under evidence/behavior/negative_control. Do not count the faulty trace as a passing DUT run or replace any of the nine mandatory runs with it. Verifier implements and executes this control after Designer handoff; Designer neither authors nor runs it in preparation.

ARCHITECTURE, PACKING AND SYNTHESIS CHECKS Validate the supplied JSON, all 413 expanded tensor names/shapes, default parameter values, packing boundaries and invalid- parameter rejection. Elaborate the default RTL with Icarus without allocating a full model or running default-size inference. Independently test BF16 packer conversion and rejection using synthetic safetensors fixtures. Synthesize configurations A and B with real Yosys read_verilog -sv, hierarchy -check, proc, opt, memory and check -assert passes; retain netlist JSON and statistics. Assert no latches, no unresolved/blackbox DUT cells, no synthesis-only inference bypass, live integrated compute/control state and substantive sequential/ arithmetic datapaths. Correctly handle escaped Yosys hdlname identifiers rather than weakening hierarchy assertions. Full-default physical PPA and timing closure are not claimed.

EXECUTION BUDGETS AND RETENTION Use architecture --timeout-s 50 with shared-runtime timeout_s 70; behavior --timeout-s 200 with shared-runtime timeout_s 220; synthesis --timeout-s 120 with shared-runtime timeout_s 140. The 20-second difference for each check is explicit startup, child cleanup and receipt-retention headroom, not additional functional execution allowance. The three outer ceilings total 430 seconds, leaving 50 seconds within the configured 480-second role limit for handoff and role overhead; this preserves the required minimum handoff reserve. No role-limit or runtime-state change is proposed. Verifier must implement phase-wide monotonic deadlines, not renew the full timeout for each child, and bound each child by the remaining phase allowance. Stop launching children when that allowance is exhausted. Cleanup and receipt finalization must remain bounded within the outer headroom; all timeouts fail acceptance. The behavior allowance includes compilation, primitive tests, all nine functional runs, comparisons, coverage and the negative control. Reuse compiled reduced-configuration executables across seeds where safe, but never reuse results or omit runs. The synthesis allowance covers both A and B, not 120 seconds per configuration. If these limits prove insufficient, retain the incomplete results and return a precise continuation request for independent recovery admission; do not weaken coverage or thresholds, fabricate completion, or silently extend a deadline.

All checks retain exact argv/cwd, tool versions, source identities, start/end times, timeout/exit status, raw stdout/stderr, comparison summaries and generated outputs in fresh project-relative check directories. Preserve failed checks and link repairs without overwriting prior evidence. Bounded advisory preparation probes, if genuinely necessary for Verifier, require new evidence/preparation/<unique-probe>/receipt.json directories and exact provenance; failed probes remain retained. Such probes do not replace shared-runtime checks. Missing python3, iverilog, vvp or yosys is an explicit prerequisite failure, not permission to install tools or skip acceptance. No check depends on a final product report.

2 Implementation and delivered sources

2.1 Production design

SystemVerilog/Verilog: 2478 physical lines; 2438 nonblank, non-comment code lines. Testbench lines are not counted as production RTL.

File	Lines	Code lines
design/ARCHITECTURE.md	--	--
design/pack_model.py	--	--
hf_embeddinggemma2_spec/1_Pooling/config.json	--	--
hf_embeddinggemma2_spec/config.json	--	--
hf_embeddinggemma2_spec/language_model_tensors.json	--	--
hf_embeddinggemma2_spec/modules.json	--	--
hf_embeddinggemma2_spec/sentence_bert_config.json	--	--
rtl/anvil_e1_address.sv	173	164
rtl/anvil_e1_attention.sv	281	278
rtl/anvil_e1_controller.sv	179	175
rtl/anvil_e1_mac.sv	34	34
rtl/anvil_e1_matvec.sv	99	99
rtl/anvil_e1_memory.sv	65	65
rtl/anvil_e1_nonlinear.sv	243	237
rtl/anvil_e1_norm.sv	177	175
rtl/anvil_e1_pool.sv	162	161
rtl/anvil_e1_rope.sv	130	130
rtl/anvil_e1_sequencer.sv	777	762
rtl/anvil_e1_top.sv	158	158

2.2 Verification

SystemVerilog/Verilog: 437 physical lines; 435 nonblank, non-comment code lines. Testbench lines are not counted as production RTL.

File	Lines	Code lines
tests/addresses.py	--	--
tests/addresses.sv	47	47
tests/architecture.py	--	--
tests/behavior.py	--	--
tests/forward.sv	275	273
tests/model.py	--	--
tests/packing.py	--	--
tests/primitives.py	--	--
tests/primitives.sv	62	62
tests/profile_address.sv	53	53
tests/profiling.py	--	--
tests/synthesis.py	--	--
tests/verify.py	--	--

A Original handoffs and execution records

A.1 Designer handoff

Updated rtl/anvil_e1_address.sv to replace runtime checkpoint scans with elaboration-time global offsets, local/full suffix prefixes and layer extents. Layer starts use floor(layer_index/6) preceding full layers; total weights size uses the same formula. Separated tensor and scratch combinational calculations while retaining shape functions, invalid-selection rejection, unsigned 128-bit arithmetic, overflow checks and existing register/cycle boundaries. Documented the implementation in design/ARCHITECTURE.md; preserved the bank-write repair, verification sources and prior evidence. Python3's existing retention helper ran the single

permitted Icarus static elaboration check successfully, without timeout: evidence/preparation/static-address-prefixes-20261014T2310/receipt.json. No functional simulation or synthesis ran. The unresolved issue is behavior-budget feasibility, not a demonstrated functional defect. Verifier must independently run address-boundary checks and fixed-work replay, then complete the unchanged functional matrix, numerical gates, scoreboard negative control and synthesis checks. This is design readiness, not acceptance or a speedup claim.

B Verification methods and results

B.1 architecture_and_packing

Command: ["python3", "tests/verify.py", "--check", "architecture", "--output", "evidence/architecture", "--timeout-s", "50"]

Declared tools: python3, iverilog, vvp

B.2 behavior

Command: ["python3", "tests/verify.py", "--check", "behavior", "--seeds", "17,89,203", "--configs", "A,B,C", "--output", "evidence/behavior", "--timeout-s", "200"]

Declared tools: python3, iverilog, vvp

B.3 synthesis

Command: ["python3", "tests/verify.py", "--check", "synthesis", "--configs", "A,B", "--output", "evidence/synthesis", "--timeout-s", "120"]

Declared tools: python3, yosys

B.4 Verifier handoff

Attempt 17 failed before verification began: .vulcan/engineering/run-0001/attempt-0017/stderr.log records FileExistsError for /work/project/evidence/architecture at Execution's exclusive directory creation. That namespace still contains 3,795,129,480 logical bytes, including seven approximately 542 MB sparse fixtures; it also repeats attempt 15's artifact-capture failure against the 1 GiB limit. The existing tests/packing.py compact-retention repair is unchanged from its successful advisory receipt, evidence/preparation/verifier-compact-header-retention-20261014T2323/receipt.json: all recorded source hashes match, children settled with exit zero, and retained artifacts total 417,254 bytes. Its summary records 65,280 finite BF16 comparisons, 256 nonfinite rejections, six malformed-header rejections, 413 default tensors and 1,450 exact packed-value comparisons. This turn used Python3 only for read-only size and provenance inspection; no source or evidence was changed and no check was rerun. The shared runtime must preserve the failed architecture namespace outside the retry output selector and provide a fresh evidence/architecture before executing the unchanged admitted command. Allowing directory reuse would overwrite prior evidence and retain oversized artifacts, so it is not a valid harness repair. Artifact capture and engineering acceptance remain unverified.

B.5 Recorded check: check:architecture_and_packing

Status: failed; exit: 0; wall seconds: 2.789510741997219; original records: .vulcan/engineering/run-0001/attempt-0015.

The following is a tail excerpt of actual tool output, not a separate correctness claim.

PASS: all implemented obligations for architecture

B.6 Recorded check: check:architecture_and_packing

Status: failed; exit: 1; wall seconds: 0.8130222420004429; original records: .vulcan/engineering/run-0001/attempt-0017.

The following is a tail excerpt of actual tool output, not a separate correctness claim.

B.7 Recorded check: check:architecture_and_packing

Status: passed; exit: 0; wall seconds: 2.783214828996279; original records: .vulcan/engineering/run-0001/attempt-0019.

The following is a tail excerpt of actual tool output, not a separate correctness claim.

PASS: all implemented obligations for architecture

B.8 Recorded check: check:behavior

Status: passed; exit: 0; wall seconds: 188.28847576500266; original records: .vulcan/engineering/run-0001/attempt-0020.

The following is a tail excerpt of actual tool output, not a separate correctness claim.

PASS: all implemented obligations for behavior

B.9 Recorded check: check:synthesis

Status: passed; exit: 0; wall seconds: 97.68590467500326; original records: .vulcan/engineering/run-0001/attempt-0021.

The following is a tail excerpt of actual tool output, not a separate correctness claim.

PASS: all implemented obligations for synthesis

C Independent acceptance and limitations

Anvil-E1 meets the admitted RTL delivery contract. I reviewed the twelve dedicated RTL modules, tensor-operation sequencing, signed arithmetic and rounding, external-memory ownership, reset/error handling, pooling and selected-channel normalization, together with the independently written references and public-interface testbench. The implementation retains the authoritative defaults and distinct local/full attention geometries; it does not substitute host-computed inference.

The recorded architecture, behavior and synthesis checks passed within their respective budgets. Evidence in evidence/architecture covers 413 tensor shapes, default elaboration, invalid parameters, address boundaries and packer conversion, including 65,280 finite BF16 values and rejection of 256 nonfinite values. All nine configuration/seed runs appear in evidence/behavior/run-ledger.json, with exact fixed-point comparisons, mathematical comparisons, saturation, stalls, errors and reset cancellation. The largest recorded nonsaturating output error is 0.000192654, below 0.02. The actual scoreboard accepted the unchanged trace and specifically rejected the one-bit-unit mutation at embedding transaction 0: expected 9532189, observed 9532190. The A/B Yosys scripts execute the required passes; their logs report zero structural problems, and the inspected checker requires integrated compute modules, sequential/arithmetic cells and no latches or unresolved cells.

I also inspected the retained negative-angle accuracy failure and its corrected result, the occupied-output-directory traceback, and the compact-fixture/output-rotation repairs. These failures are not counted as passing runs; the subsequent admitted checks supply acceptance evidence. Review was read-only; I executed no commands. Acceptance covers the documented synthetic quantized forward contract and reduced-configuration synthesis, not official pretrained-checkpoint parity, full-default inference, throughput, physical PPA or timing closure. No required repair remains.

The full design-input and verification-input SHA-256 manifest is retained beside this report. The PDF is a historical snapshot: later source changes require explicit revalidation and do not update this PDF. Tool and role durations include waiting; they must not be interpreted as model inference utilization.

D Engineering timeline and retained failures

Role / operation	UTC start	Seconds	Status
coordinator / plan	21:27:55	275.57	passed
reviewer / plan	21:32:31	47.44	passed
coordinator / plan	21:33:19	116.07	passed
reviewer / plan	21:35:16	33.93	passed
designer / designer:6	21:35:52	424.34	passed
coordinator / route	21:43:00	46.56	passed
reviewer / recovery	21:43:49	64.73	passed
designer / designer:9	21:44:57	447.84	passed
coordinator / route	21:52:30	37.30	passed
reviewer / recovery	21:53:11	49.45	passed
designer / designer:12	21:54:06	480.13	failed
coordinator / route	22:02:13	50.47	passed
reviewer / recovery	22:03:08	62.96	passed
verifier / verifier:15	22:04:16	164.13	passed
designer / designer:16	22:07:06	69.36	passed
verifier / verifier:17	22:08:20	459.96	passed
coordinator / route	22:16:09	66.62	passed
reviewer / recovery	22:17:24	60.16	passed
verifier / verifier:20	22:18:32	443.66	passed
coordinator / route	22:26:07	48.03	passed
reviewer / recovery	22:27:02	72.67	passed
verifier / verifier:23	22:28:23	469.92	passed
coordinator / route	22:36:25	92.19	passed
reviewer / recovery	22:38:05	75.96	passed
verifier / verifier:26	22:39:31	468.74	passed
coordinator / route	22:47:31	49.76	passed
reviewer / recovery	22:48:28	60.04	passed
verifier / verifier:29	22:49:38	477.99	passed
designer / designer:30	22:57:43	133.75	passed
verifier / verifier:31	23:00:05	479.68	passed
coordinator / route	23:08:16	45.40	passed
reviewer / recovery	23:09:09	57.49	passed
designer / designer:34	23:10:17	141.85	passed
verifier / verifier:35	23:12:46	439.62	passed
tool execution / check:architecture_and_packing	23:20:13	2.79	failed
verifier / verifier:37	23:20:24	147.36	passed
tool execution / check:architecture_and_packing	23:22:58	0.81	failed
verifier / verifier:39	23:23:08	62.82	passed
coordinator / route	23:24:21	29.32	passed
reviewer / recovery	23:24:58	53.55	passed
tool execution / check:architecture_and_packing	23:31:13	2.78	passed
tool execution / check:behavior	23:31:22	188.29	passed
tool execution / check:synthesis	23:34:36	97.69	passed
reviewer / review	23:36:23	147.95	passed

run-0001/failure-0001 Current contract.

{"id": "failure-0001", "step": "designer:12", "status": "resolved", "attempts": [3], "cause": "unknown", "detail": "timeout", "diagnoses": [], "recovery_checks": {"architecture_and_packing": 19, "behavior": 20, "synthesis": 21}, "review_attempt": 22}

run-0001/failure-0002 Current contract.

{"id": "failure-0002", "step": "check:architecture_and_packing", "status": "resolved", "attempts": [15, 17], "cause": "unknown", "detail": "nonzero_exit", "diagnoses": [{"cause": "unknown", "note": "Diagnosed architecture_and_packing: the command exited 0, but capture failed because seven sparse safetensors fixtures contributed approximately 3.79 GB of logical artifacts, exceeding the 1 GiB limit. Updated only tests/packing.py to retain exact header bytes, hashes, logical extents and reconstruction metadata; identical full-size sparse files are reconstructed temporarily for the unchanged header checks. The complete reduced packer CLI fixture remains retained. Bounded repair probe passed: evidence/preparation/verifier-compact-header-retention-20261014T2323/receipt.json, linked to evidence/architecture/receipt.json. Python3 compared 65,280 finite BF16 values, rejected 256 nonfinite values and six malformed headers, validated 413 default tensors and exactly compared 1,450 packed values. Retained probe artifacts total 417,254 bytes. No RTL, design inputs, acceptance thresholds, admitted commands or prior evidence changed; no full acceptance suite was rerun. Before retrying the unchanged admitted command, the shared runtime must preserve the failed evidence/architecture namespace outside the fresh retry output selector; otherwise stale oversized artifacts remain. Artifact capture and delivery acceptance remain unverified.", "repair": 1}, {"cause": "unknown", "note": "Retired occupied evidence/architecture namespace and updated tests/verify.py to rotate prior output directories", "repair": 2}], "pass_attempt": 19, "review_attempt": 22}

E Professional knowledge and provenance

E.1 coordinator

Vulcan: 0.7.0; domain: digital_circuit; delivery: explicit_prompt_context. These records prove what context was supplied, not that every method was correctly applied.

assets/skills/coordinator/SKILL.md
SHA-256 d0a7243c90830bc1e6243d6bca62c9b5731ce9affbfac5bdb015d11a51d4d14e
argus_verticals/digital_circuit/skills/engineer/digital-knowledge-map.md
SHA-256 6a7ad544f79398b80b87922a9121e0ecb0351af278d0776e9bf03c404564eb9c

E.2 reviewer

Vulcan: 0.7.0; domain: digital_circuit; delivery: explicit_prompt_context. These records prove what context was supplied, not that every method was correctly applied.

assets/skills/reviewer/SKILL.md
SHA-256 4579bbde560e92bea30171290e2448bbacac042ee5e4d211c13110ffd712962e
argus_verticals/digital_circuit/skills/reviewer/digital-circuit-signoff-review.md
SHA-256 e74b8c8d7acdc24ade024f7bfcc8fecc4763e0321ba9e9e5d3c505d13ce5bc6a
argus_verticals/digital_circuit/verification/skills/reviewer/verification-acceptance.md
SHA-256 6dffec449aaec0d01762b3f1d64ed9e11869369f069245e44a0a672b76c324e2

E.3 designer

Vulcan: 0.7.0; domain: digital_circuit; delivery: explicit_prompt_context. These records prove what context was supplied, not that every method was correctly applied.

assets/skills/designer/SKILL.md
SHA-256 c2e5f26720cb23f63bbee67f77a0487b21fb47393f0b0145d518b20d14ab2d18
argus_verticals/digital_circuit/skills/engineer/digital-knowledge-map.md
SHA-256 6a7ad544f79398b80b87922a9121e0ecb0351af278d0776e9bf03c404564eb9c

E.4 verifier

Vulcan: 0.7.0; domain: digital_circuit; delivery: explicit_prompt_context. These records prove what context was supplied, not that every method was correctly applied.

assets/skills/verifier/SKILL.md
SHA-256 86b3b0305702612aaf354cb637a26a15fb2fc9b546de3d3a93b2c71beecaf6d3
argus_verticals/digital_circuit/verification/skills/engineer/verification-matrix.md
SHA-256 32f7f6b25d834cec897239e601a1417c17d034eb27d9a117321c8d7782ec28bf
argus_verticals/digital_circuit/verification/skills/engineer/protocol-verification.md
SHA-256 5b1ed82543ff18472b403c4b44fdfd0786f4e7a6ec0f65c17937642992062244

argus: 1d5ee00a02e2d400443eed925e6079909f8f4e25
argus-verticals: 73eecd72e1d6a4c530fca0a379576b5554fb5eec